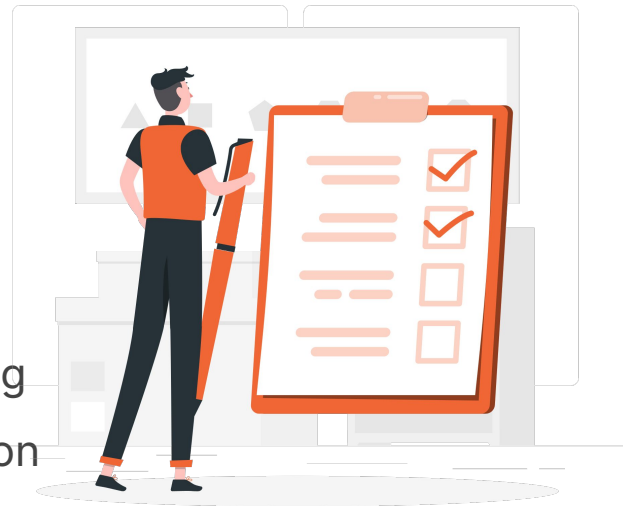




Azure OpenAI and AI Search

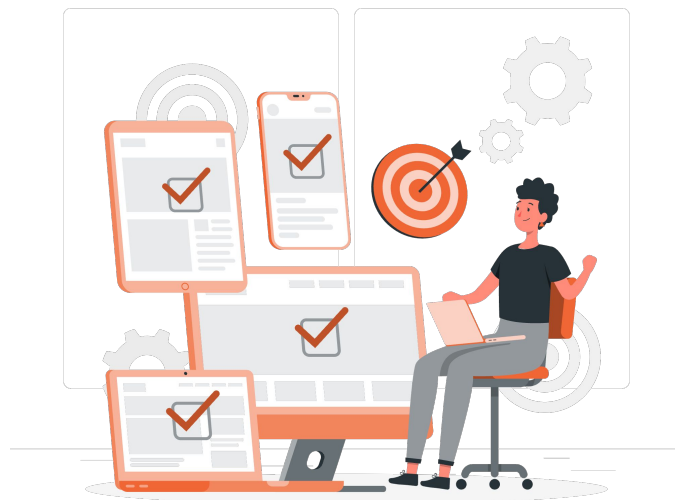
Lesson

1. Azure OpenAI overview
2. Azure OpenAI model deployment
3. Azure OpenAI quotas & rate limits
4. Azure AI Search overview
5. Azure AI Search data sources & indexing
6. Building Retrieval-Augmented Generation



Learning Outcome

1. Deploy and manage models in Azure OpenAI
2. Secure access using API Keys and RBAC
3. Build and index data for AI Search
4. Improve retrieval with semantic ranking
5. Implement RAG using OpenAI and AI Search



Azure OpenAI Overview



Azure OpenAI Overview

Azure OpenAI is a cloud service from Microsoft that enables the use of advanced AI models such as those from OpenAI, hosted securely on the Microsoft Azure platform.



Feature	OpenAI	Azure OpenAI
Infrastructure	Managed by OpenAI independently	Managed by Microsoft (Azure)
Data Privacy	Customer data used for training	No customer data used for training
Compliance	OpenAI Enterprise standard	Full Azure compliance (HIPAA, SOC2)
Support/SLA	Platform support	Enterprise-grade SLAs
Security	Standard API security	VNETs, RBAC, Private Links

Azure OpenAI Fundamentals

Core platform idea

Azure OpenAI gives organizations access to OpenAI models through Azure-managed deployments. Microsoft naming has evolved through Foundry and classic experiences, but the durable workflow is stable: choose the resource context, pick a model, create a deployment, and have the application call that deployment.

1 Resource

The Azure or Foundry resource is the service boundary that owns the deployment.

2 Model

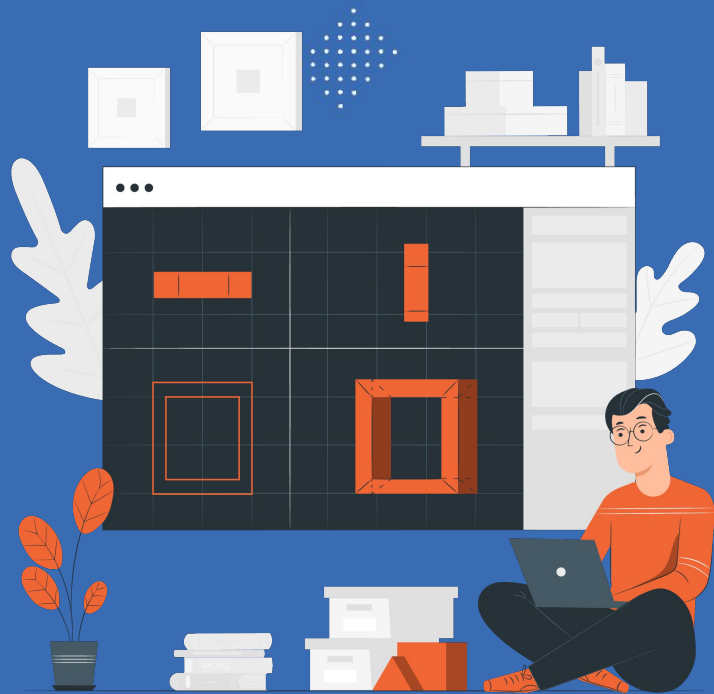
The model family is what you choose to serve, not what the app calls directly.

3 Deployment

The deployment is the runtime identity the application uses.

Key point: the deployment is not an administrative afterthought. It is the app-facing object that links model choice to real runtime behavior.

Azure OpenAI Model Deployment



Model Name vs Deployment Name

Common first assumption

If the chosen model is `gpt-4o` or another model family, the code should call that model name directly. This assumption comes from non-Azure examples and misses the Azure deployment layer.

What Azure applications actually target

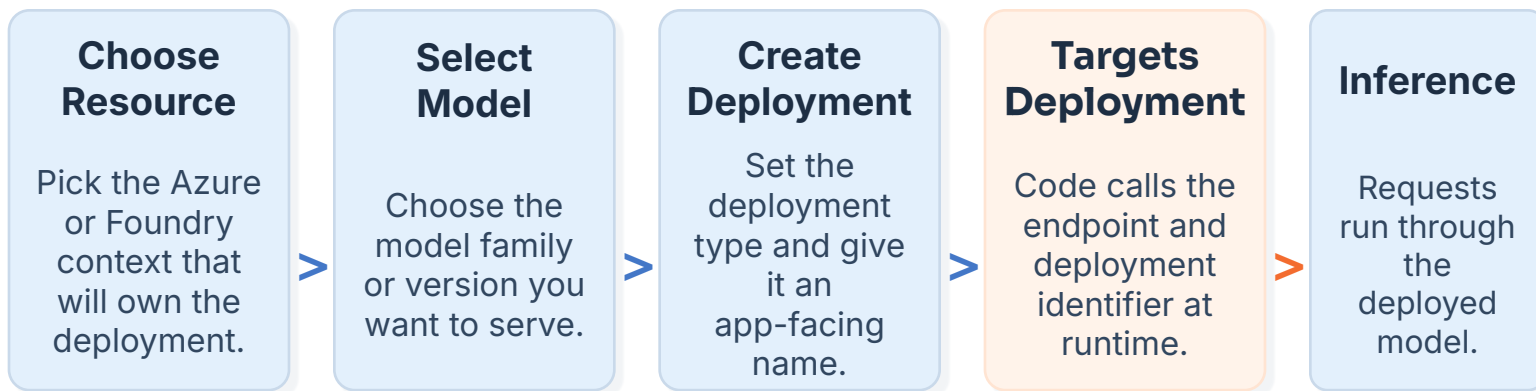
The application points to the Azure endpoint and the deployment identifier that you created. Clear deployment names make code, debugging, and team communication easier.

Application view of the runtime target

Azure endpoint
+
Deployment name
=
The model-serving address your code depends on

Operational consequence: rename carelessly, and every environment, secret, config reference, and troubleshooting note becomes harder to follow. Deployment naming is part of solution design, not just portal hygiene.

Deployment Concept Diagram



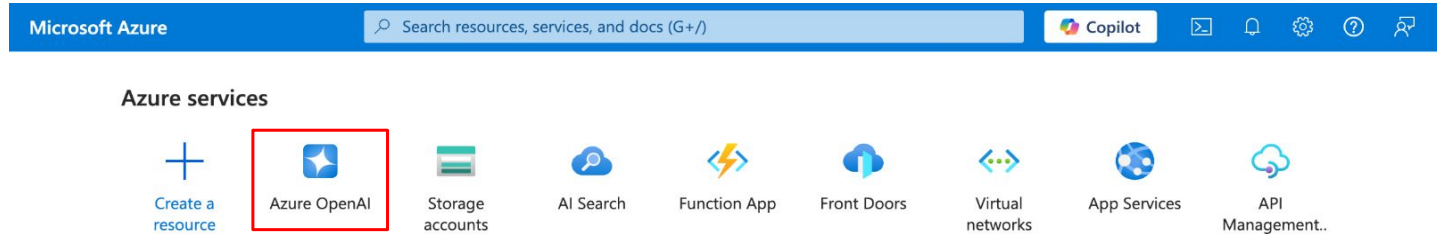
This is the core concept that makes later topics click: quotas apply to real deployments, lifecycle events affect real deployments, and authentication protects access to real deployments.



Let's Create Azure OpenAI Instance

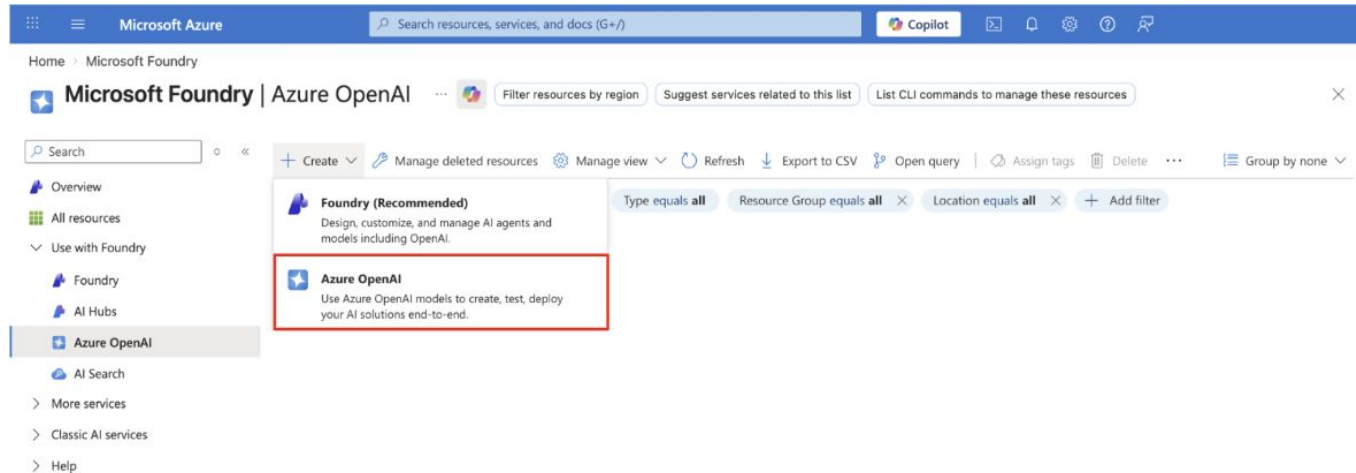
Azure OpenAI: Portal

- Visit URL <https://portal.azure.com/>
- Login with your email and password
- It will open the azure portal
- Click **Azure OpenAI**



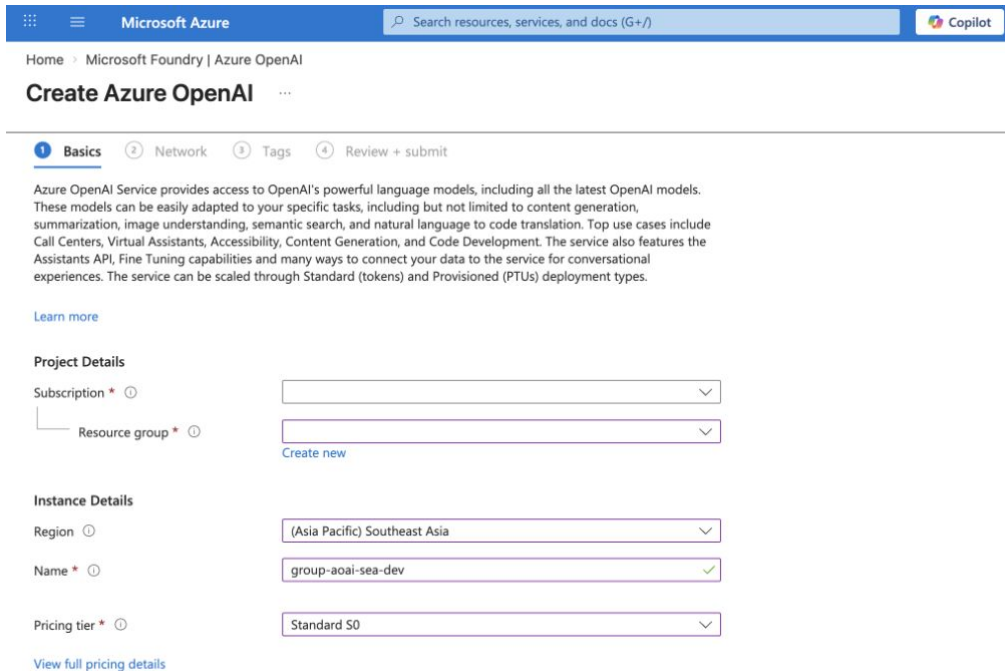
Azure OpenAI: Create

- It will open the Microsoft Foundry or Azure OpenAI page.
- To create instance, click **Create**.
- Then choose **Azure OpenAI**.



Azure OpenAI: Basics

- Fill the project details in basics section.
- Choose your subscription, resource group, and region based on your company's policy.
- For the name, use lowercase letter with format:
{group}-{instance}-{region}-{environment}.
For example: bsi-aoai-sea-dev
- Choose standard pricing tier
- Then, click **Next**.



Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home > Microsoft Foundry | Azure OpenAI

Create Azure OpenAI

1 Basics 2 Network 3 Tags 4 Review + submit

Azure OpenAI Service provides access to OpenAI's powerful language models, including all the latest OpenAI models. These models can be easily adapted to your specific tasks, including but not limited to content generation, summarization, image understanding, semantic search, and natural language to code translation. Top use cases include Call Centers, Virtual Assistants, Accessibility, Content Generation, and Code Development. The service also features the Assistants API, Fine Tuning capabilities and many ways to connect your data to the service for conversational experiences. The service can be scaled through Standard (tokens) and Provisioned (PTUs) deployment types.

[Learn more](#)

Project Details

Subscription * ⓘ

Resource group * ⓘ

[Create new](#)

Instance Details

Region ⓘ

Name * ⓘ

Pricing tier * ⓘ

[View full pricing details](#)

Azure OpenAI: Pricing

Region:

Southeast Asia

Currency:

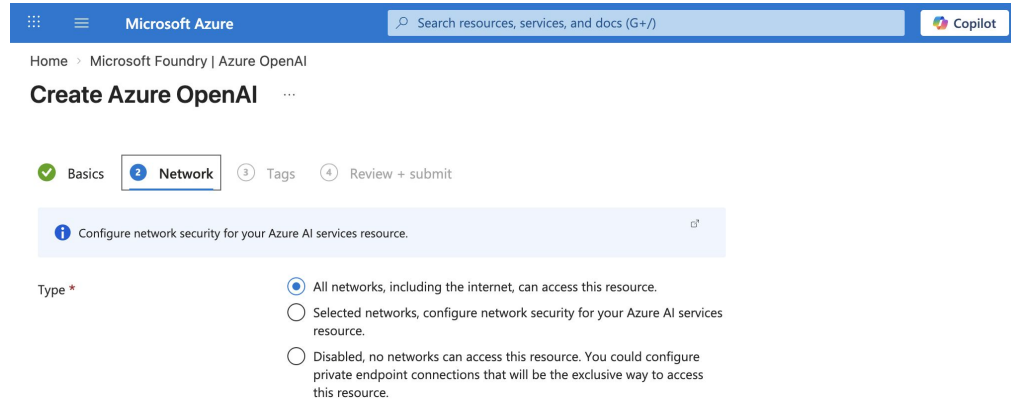
United States – Dollar (\$) USD

GPT 5 Series

GPT-5 Pro Global	Input: N/A Output: N/A	N/A	N/A
GPT-5 Codex Global	Input: N/A Cached Input: N/A Output: N/A	N/A	N/A
GPT-5-mini Global	Input: \$0.25 Cached Input: \$0.03 Output: \$2	N/A	N/A
GPT-5-mini Data Zone	Input: N/A Cached Input: N/A Output: N/A	N/A	N/A
GPT-5-nano Global	Input: \$0.05 Cached Input: \$0.01 Output: \$0.40	N/A	N/A

Azure OpenAI: Network

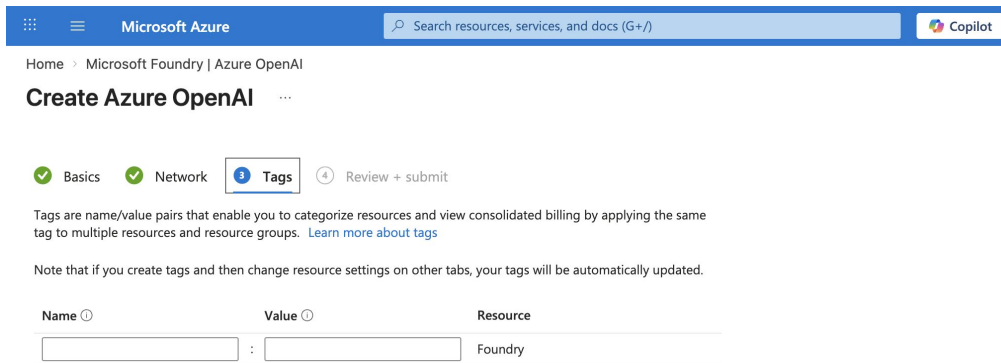
- Configure the network security in network section.
- Choose all networks if there are no customize network security for your service.
- Then, click **Next**.



The screenshot shows the Microsoft Azure portal interface for creating an Azure OpenAI resource. The top navigation bar includes the Microsoft Azure logo, a search bar, and the Copilot icon. The breadcrumb trail indicates the path: Home > Microsoft Foundry | Azure OpenAI. The main heading is 'Create Azure OpenAI'. Below this, there are four steps in a sequence: 1. Basics (completed), 2. Network (current step), 3. Tags, and 4. Review + submit. A light blue information box contains the text: 'Configure network security for your Azure AI services resource.' Below this, the 'Type' section has three radio button options: 'All networks, including the internet, can access this resource.' (selected), 'Selected networks, configure network security for your Azure AI services resource.', and 'Disabled, no networks can access this resource. You could configure private endpoint connections that will be the exclusive way to access this resource.'

Azure OpenAI: Tags

- Configure the name/value in tags section.
- Tags are name/value pairs that enable you to categorize resource.
- Then, click **Next**.



The screenshot shows the 'Create Azure OpenAI' wizard in the Microsoft Azure portal. The 'Tags' step is selected and highlighted with a red box. The wizard has four steps: Basics, Network, Tags, and Review + submit. Below the steps, there is a description of tags and a note about automatic updates. At the bottom, there is a table with three columns: Name, Value, and Resource. The 'Resource' column contains the value 'Foundry'.

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home > Microsoft Foundry | Azure OpenAI

Create Azure OpenAI

Basics Network **Tags** Review + submit

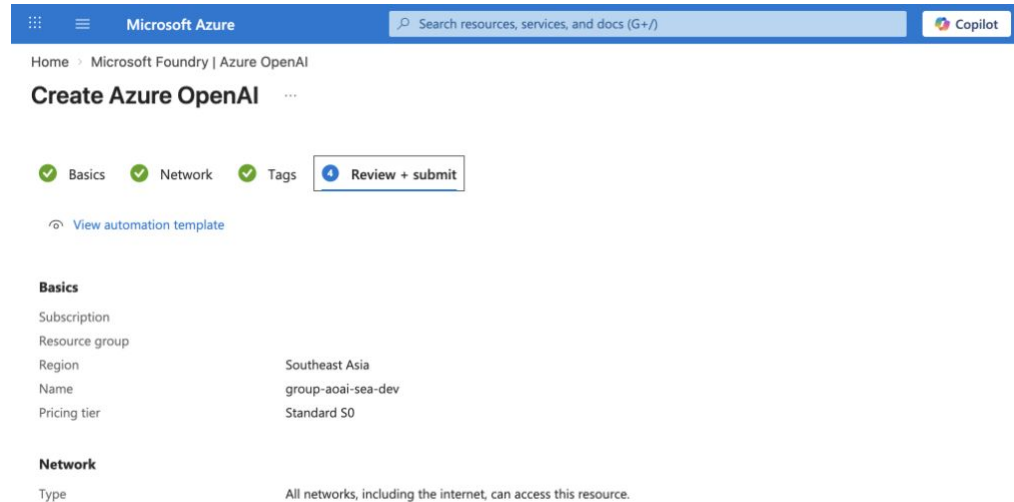
Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more about tags](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name	Value	Resource
	:	Foundry

Azure OpenAI: Review

- Review the basics and network settings that have already been configured before submitting.
- If everything is correct, click **Submit**.



Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home > Microsoft Foundry | Azure OpenAI

Create Azure OpenAI

✓ Basics ✓ Network ✓ Tags **Review + submit**

[View automation template](#)

Basics

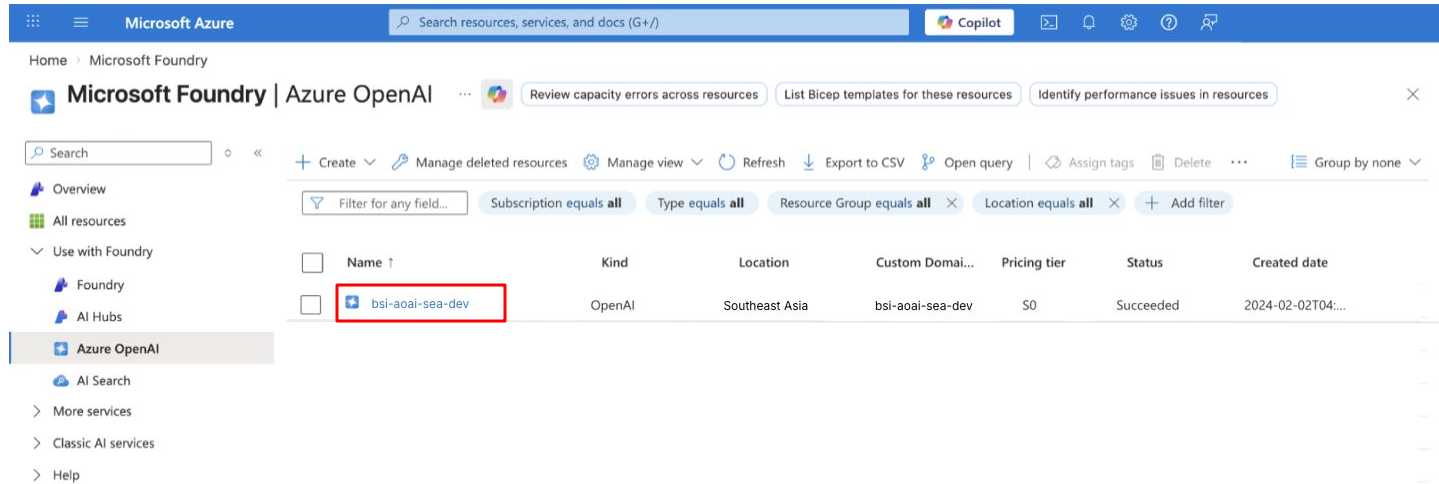
Subscription	
Resource group	
Region	Southeast Asia
Name	group-aoai-sea-dev
Pricing tier	Standard S0

Network

Type	All networks, including the internet, can access this resource.
------	---

Azure OpenAI: Get the API Key

- Open the Microsoft Foundry or Azure OpenAI page again.
- Then, click the instance that was created.

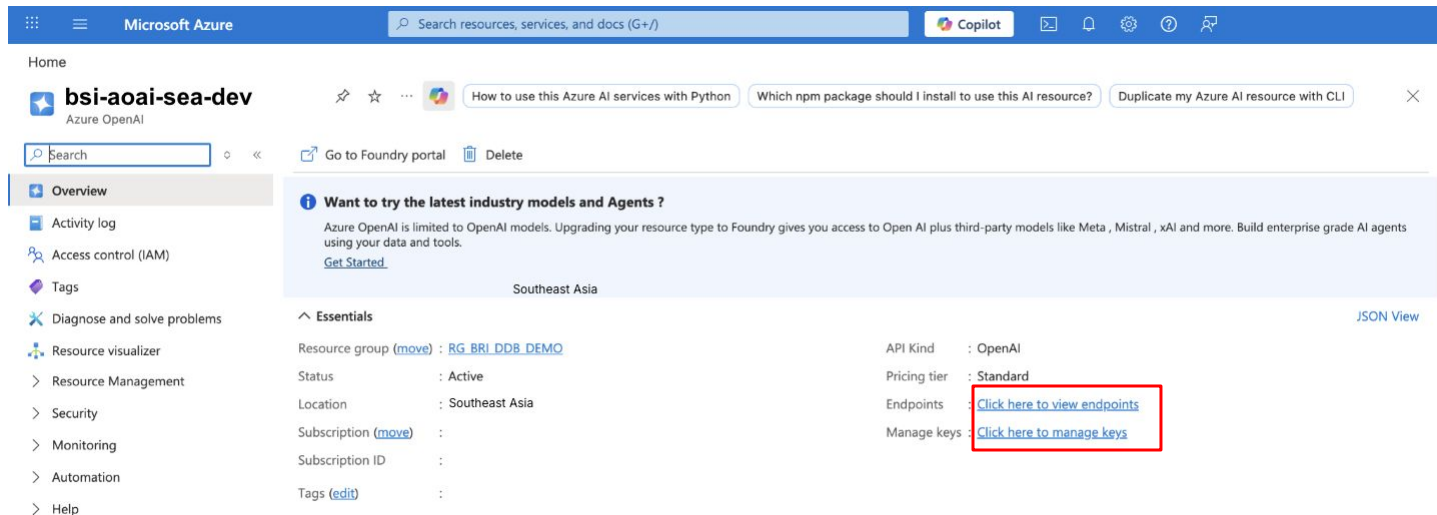


The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and the Copilot icon. The main header indicates the current location is 'Home > Microsoft Foundry'. Below this, the 'Microsoft Foundry | Azure OpenAI' section is active, displaying three tabs: 'Review capacity errors across resources', 'List Bicep templates for these resources', and 'Identify performance issues in resources'. The left sidebar contains a navigation menu with options like 'Overview', 'All resources', 'Use with Foundry', 'Foundry', 'AI Hubs', 'Azure OpenAI' (which is selected), 'AI Search', 'More services', 'Classic AI services', and 'Help'. The main content area shows a table of Azure OpenAI instances. The table has columns for 'Name', 'Kind', 'Location', 'Custom Domain', 'Pricing tier', 'Status', and 'Created date'. A red box highlights the instance named 'bsi-aai-sea-dev'.

☐	Name ↑	Kind	Location	Custom Domain...	Pricing tier	Status	Created date
☐	bsi-aai-sea-dev	OpenAI	Southeast Asia	bsi-aai-sea-dev	S0	Succeeded	2024-02-02T04:...

Azure OpenAI: Manage Keys

- It will open the Azure OpenAI Instance page.
- Then, click the **Endpoints** or **Manage keys**



Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home

bsi-aoai-sea-dev
Azure OpenAI

How to use this Azure AI services with Python | Which npm package should I install to use this AI resource? | Duplicate my Azure AI resource with CLI

Search

Go to Foundry portal | Delete

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Resource Management
- Security
- Monitoring
- Automation
- Help

Want to try the latest industry models and Agents ?

Azure OpenAI is limited to OpenAI models. Upgrading your resource type to Foundry gives you access to Open AI plus third-party models like Meta , Mistral , xAI and more. Build enterprise grade AI agents using your data and tools.
[Get Started](#)

Southeast Asia

Essentials

Resource group (move) : [RG_BRI_DDB_DEMO](#)

Status : Active

Location : Southeast Asia

Subscription (move) :

Subscription ID :

Tags (edit) :

API Kind : OpenAI

Pricing tier : Standard

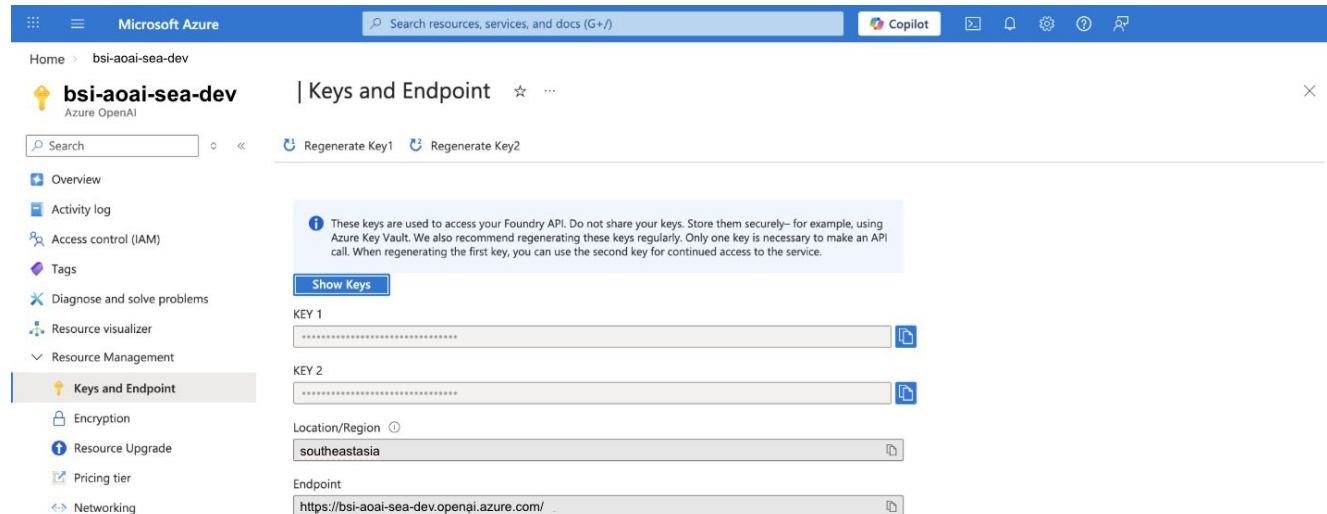
Endpoints : [Click here to view endpoints](#)

Manage keys : [Click here to manage keys](#)

JSON View

Azure OpenAI: Keys and Endpoint

- It will open the keys and endpoint section.
- Click the copy icon in key 1 or key 2 for the **API Key**.
- Click the copy icon in endpoint for the **Endpoint**.



The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and the Copilot icon. The breadcrumb trail indicates the current location: Home > bsi-aoai-sea-dev. The main content area is titled 'Keys and Endpoint' and contains a warning message about key security, a 'Show Keys' button, and fields for KEY 1, KEY 2, Location/Region (set to 'southeastasia'), and Endpoint (set to 'https://bsi-aoai-sea-dev.openai.azure.com/'). Each field has a copy icon to its right. The left sidebar shows the resource 'bsi-aoai-sea-dev' and a list of navigation options, with 'Keys and Endpoint' currently selected.

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > bsi-aoai-sea-dev

bsi-aoai-sea-dev
Azure OpenAI

Keys and Endpoint

Search

Regenerate Key1 Regenerate Key2

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Resource Management
Keys and Endpoint
Encryption
Resource Upgrade
Pricing tier
Networking

These keys are used to access your Foundry API. Do not share your keys. Store them securely—for example, using Azure Key Vault. We also recommend regenerating these keys regularly. Only one key is necessary to make an API call. When regenerating the first key, you can use the second key for continued access to the service.

Show Keys

KEY 1

KEY 2

Location/Region
southeastasia

Endpoint
https://bsi-aoai-sea-dev.openai.azure.com/

Azure OpenAI Quotas and Rate Limits



Quotas and Rate Limits

First principle

A successful deployment is not proof that the system is ready for production traffic. Azure OpenAI quota and rate behavior shape how much real traffic a deployment can serve. Understand the policy shape first, because the exact numbers can change.

1 Subscription

Quota scope includes the subscription context.

2 Region

Capacity is also tied to regional allocation.

3 Model

Different models can behave under different limits.

4 Type

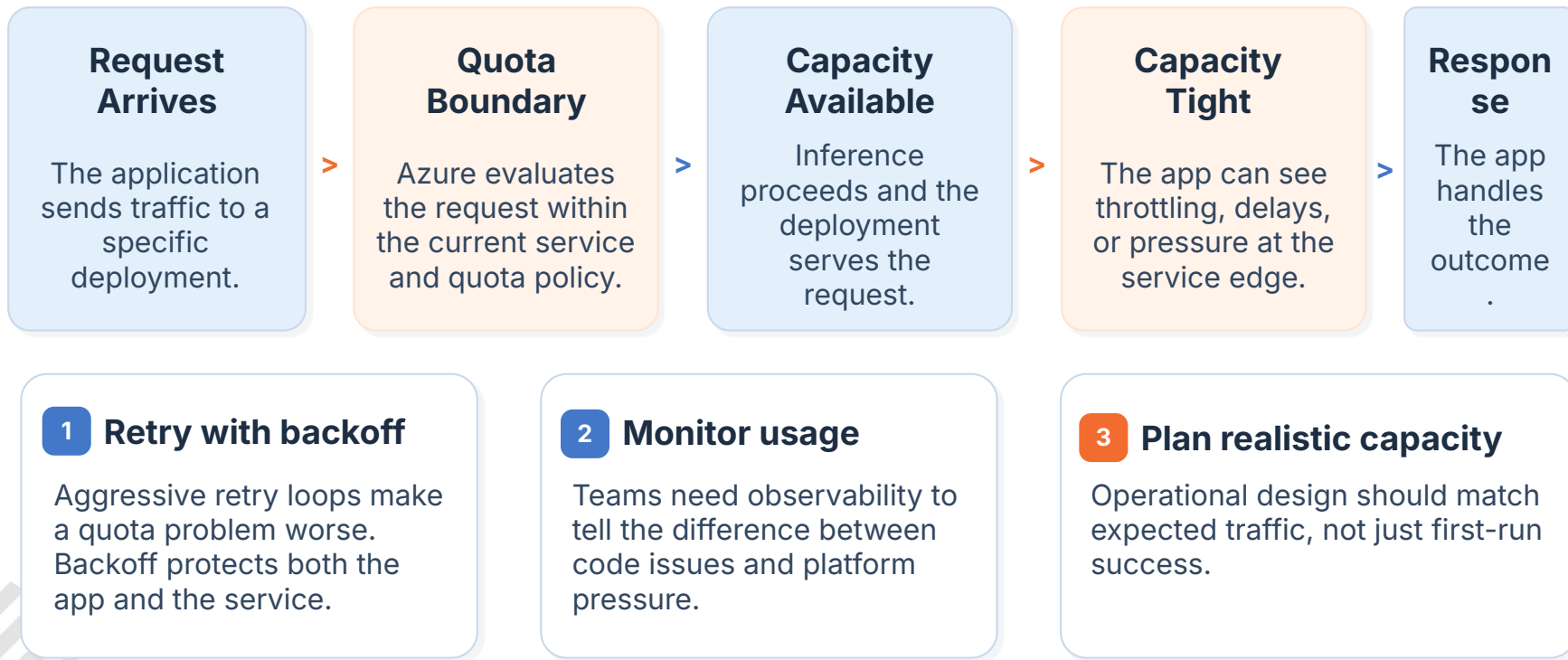
Deployment type can change throughput characteristics.

Quota Guidance

We are introducing Quota Tiers to improve the Foundry Models experience and reduce friction

Use this screenshot to identify where current official scope and allocation rules are published instead of memorizing stale values.

Quota and Rate-Limit Operational



Capacity Planning & Throttling Responses

Symptom: requests slow down

Interpret this as a signal to inspect service behavior and traffic shape, not only application code.

1 Inspect the current quota scope

Quota rules are time-sensitive. Verify the current Microsoft guidance.

2 Review deployment characteristics

Not all models or deployment types carry identical throughput expectations.

Symptom: throttling appears

The platform may be protecting its service boundary when the app sends too much traffic too fast.

3 Design app-side resilience

Retries, backoff, and monitoring belong in production design.

4 Plan before scale events

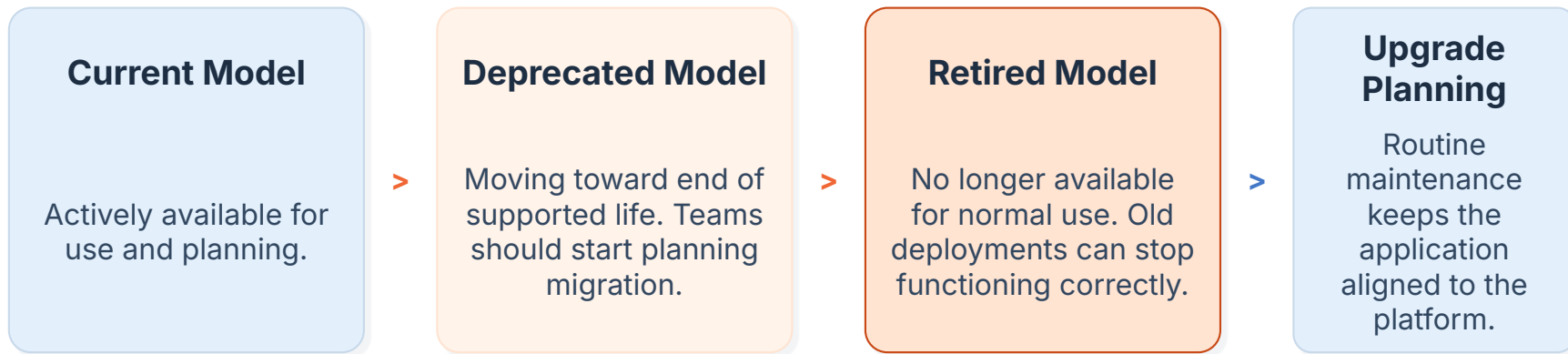
Capacity planning should happen before demand spikes, not after failures.

Symptom: two deployments behave differently

Check region, model, and deployment type before assuming a coding regression.

Operational insight: a platform limit surfacing in the app is still part of application behavior. Good engineering treats quotas as design constraints, not as invisible infrastructure.

Model Lifecycle Overview



Operational lesson: choosing a model is not a one-time event. Teams need a habit of checking availability, deprecation notices, and retirement timelines as part of normal platform maintenance.

Deprecated vs Retired

Deprecated

The model is moving toward the end of its supported life. It may still be usable under specific conditions, but teams should treat it as a migration warning and start planning the next supported option.

Retired

The model is no longer available for normal use. If a team ignores retirement guidance, application behavior can fail because the underlying deployment is no longer supported.

1 Monitor notices

Lifecycle review belongs in regular platform checks.

2 Plan upgrade paths

Do not wait for runtime failures before moving to supported models.

3 Test before cutover

Treat model upgrades as operational changes, not just label swaps.

Authentication

Two common access patterns

API keys are easy to visualize and good for first understanding the request flow. Microsoft Entra ID based access, often described as keyless authentication, aligns better with enterprise security because access can be attached to identities and roles rather than shared secrets.

API Key

Fastest way to explain the mechanics of access.



Entra ID

Identity-based access that removes the need to distribute keys.



RBAC

Roles determine whether a given identity can call the service.

Keys are easier to start with, but identity and role-based access provide the stronger long-term security posture.

API Keys vs Entra ID

API keys

Best for first-run understanding and simple demos.

Strength: easy to map into app configuration.

Risk: keys can be copied, exposed, or managed too broadly if teams are careless.

Practical point: convenience comes with a security tradeoff.

Microsoft Entra ID plus RBAC

Best for enterprise posture.

Strength: access is attached to identities and permissions instead of shared secrets.

Risk: the identity and role chain is less intuitive at first and needs careful setup.

Practical point: this is the recommended direction where feasible.

Azure AI Search Overview



Azure AI Search Overview

Azure AI Search is an information retrieval and search platform, designed to optimize retrieval-augmented generation (RAG) within Gen AI applications.



[Home](#) >

Marketplace

Get Started

Service Providers

Management

Private Marketplace

Private Offer Management

My Marketplace

Favorites

My solutions

Recently created

Private plans

Categories

AI + Machine Learning (137)

☐ Azure benefit eligible only ☐ Azure services only

Showing 1 to 20 of 207 results for 'Azure AI Search'. [Clear search](#)



Azure AI Search

Microsoft

Azure Service

AI-powered cloud search service for mobile and web app development (formerly Azure Cognitive Search)

Create

AJSERA

AI Search

Aisera

SaaS

LLM-powered search for personalized, accurate, & permission-aware results across all data sources

Price varies

Subscribe



Elastic Search (Elasticsearch) – An Azure Native ISV

Elastic

SaaS

The Elastic Search AI Platform brings together the precision of search and the intelligence of AI.

Price varies

Subscribe

BA Insight

Free trial

BA Insight for Azure AI Search

BA-Insight - Global HQ (Boston)

SaaS

Creating relevant, personalized, actionable, and intelligent AI content search is easier than ever.

Price varies

Subscribe



Evoke Workspace Search - EWS

Evoke Technologies Private Limit...

Azure Application

EWS is an AI-powered platform that uses GenAI combining large language models with enterprise data.

Starts at
\$249.00/month

Create

Azure AI Search Fundamentals

Retrieval layer for enterprise AI

Azure AI Search helps an application find relevant content before generation. If Azure OpenAI writes the answer, Azure AI Search helps the system decide which content should be considered first. This matters because internal organizational knowledge must be ingested, indexed, and queried deliberately.

1 Index

Structured searchable content with fields and search behavior.

2 Indexer

Automated pull-based ingestion from supported sources.

3 Query

Retrieval happens later, at query time when the app asks for results.

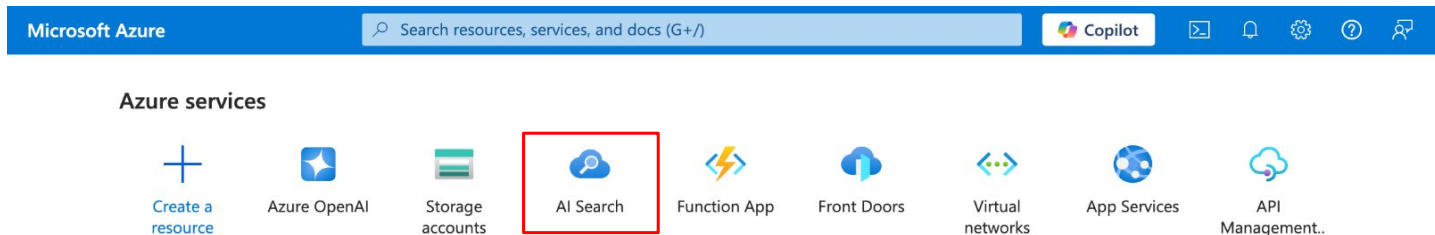
Key distinction: indexing prepares content. Query-time retrieval consumes that prepared content. Search quality does not begin at the query box.



Let's Create Azure AI Search Instance

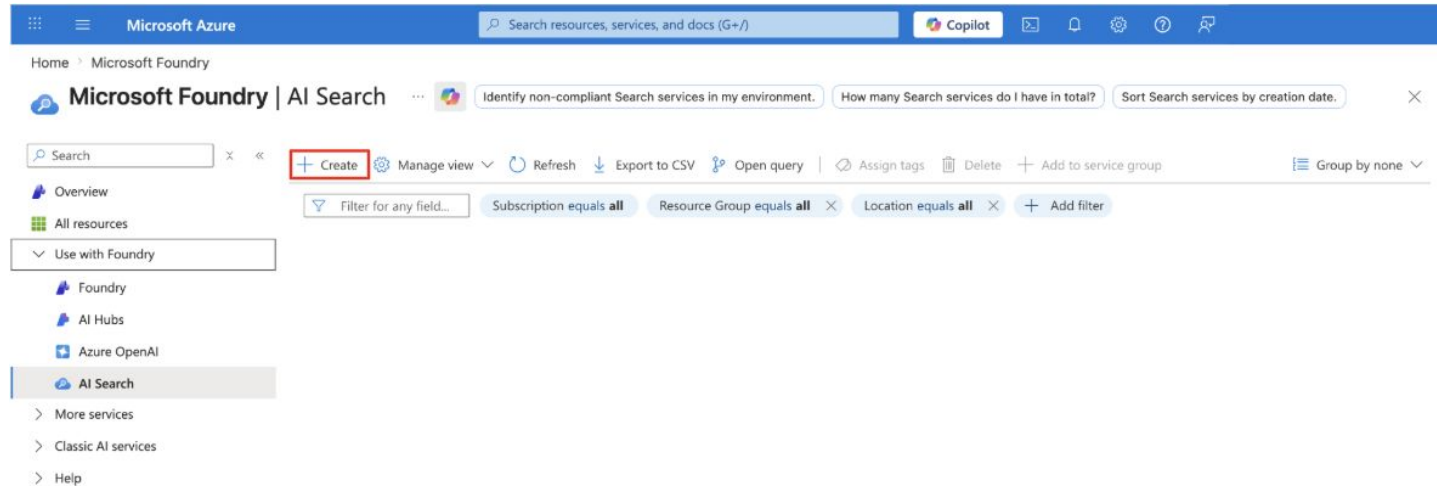
Azure AI Search: Portal

- Visit URL <https://portal.azure.com/>
- Login with your email and password
- It will open the azure portal
- Click **Azure AI Search**



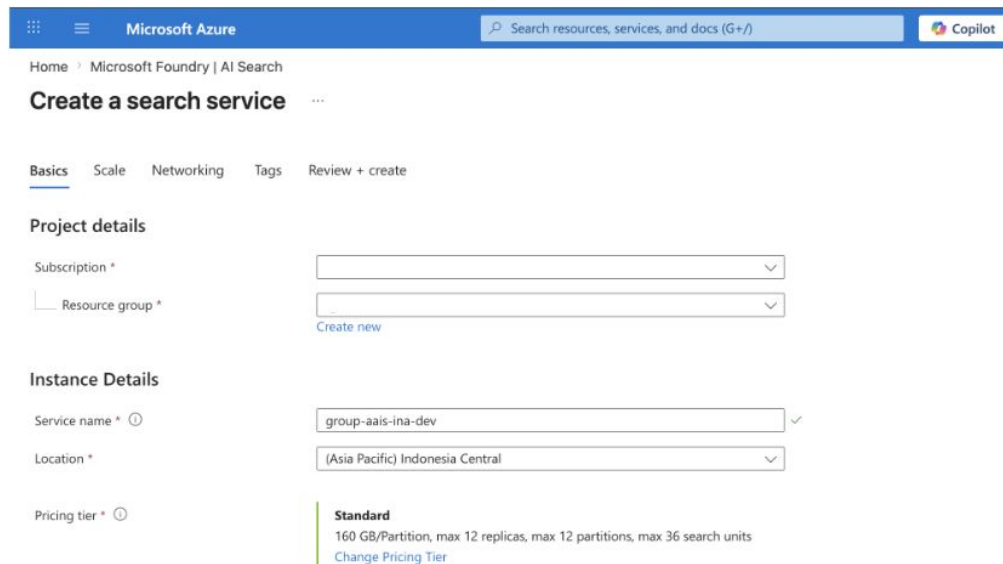
Azure AI Search: Create

- It will open the Microsoft Foundry or Azure AI Search page.
- To create instance, click **Create**.



Azure AI Search: Basics

- Fill the project details in basics section.
- Choose your subscription, resource group, and location based on your company's policy.
- For the name, use lowercase letter with format:
{group}-{instance}-
{region}-{environment}.
For example: bsi-aais-ina-dev
- Choose standard pricing tier
- Then, click **Next**.



The screenshot shows the 'Create a search service' page in the Microsoft Azure portal, specifically the 'Basics' tab. The page is titled 'Create a search service' and has a breadcrumb trail 'Home > Microsoft Foundry | AI Search'. Below the title, there are tabs for 'Basics', 'Scale', 'Networking', 'Tags', and 'Review + create'. The 'Basics' tab is selected. The 'Project details' section contains two dropdown menus: 'Subscription *' and 'Resource group *'. Below the 'Resource group *' dropdown is a link that says 'Create new'. The 'Instance Details' section contains three fields: 'Service name *' with a help icon, 'Location *', and 'Pricing tier *' with a help icon. The 'Service name *' field contains the text 'group-aais-ina-dev' and has a checkmark on the right. The 'Location *' dropdown is set to '(Asia Pacific) Indonesia Central'. The 'Pricing tier *' dropdown is set to 'Standard', which is highlighted with a green border. Below the 'Standard' tier, the text reads '160 GB/Partition, max 12 replicas, max 12 partitions, max 36 search units' and there is a link 'Change Pricing Tier'.

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home > Microsoft Foundry | AI Search

Create a search service

Basics Scale Networking Tags Review + create

Project details

Subscription *

Resource group *

Create new

Instance Details

Service name * ⓘ group-aais-ina-dev ✓

Location * (Asia Pacific) Indonesia Central

Pricing tier * ⓘ

Standard
160 GB/Partition, max 12 replicas, max 12 partitions, max 36 search units
[Change Pricing Tier](#)

Azure AI Search: Pricing

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Microsoft Foundry | AI Search > Create a search service

Select Pricing Tier

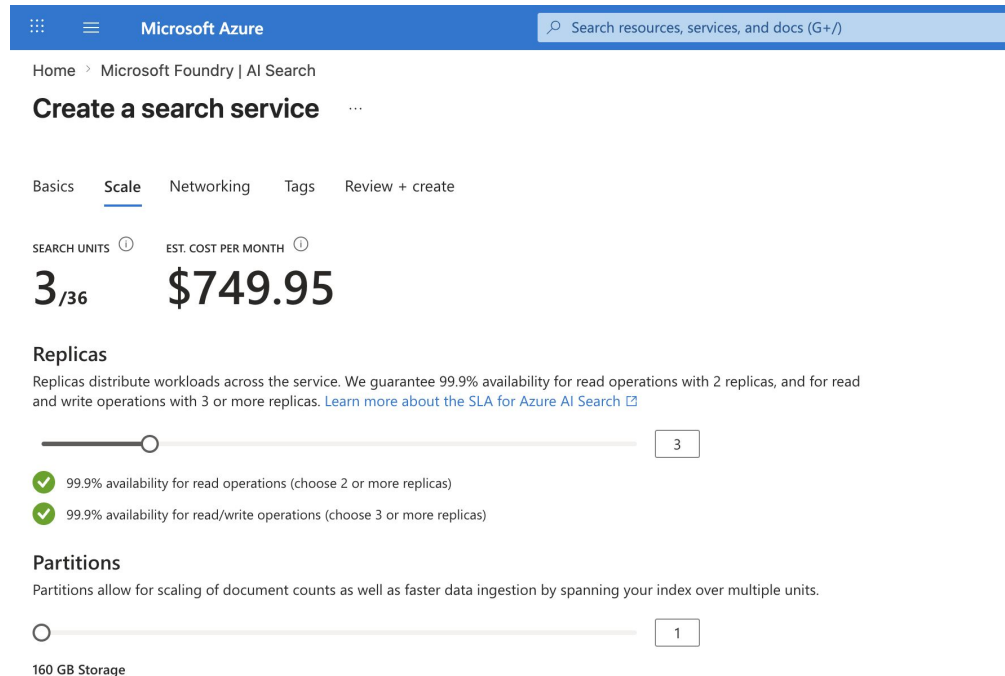
Browse available skus and their features

SKU	Offering	Indexes	Indexers	Vector quota	Total storage	Search ...	Replicas	Partitio...	Search unit cost...
F	Free	3	3	25 MB	50 MB	1	1	1	\$0.00
B	Basic	15	15	5 GB/Partition	15 GB/Partition	9	3	3	\$75.14
S	Standard	50	50	35 GB/Partition	160 GB/Partition	36	12	12	\$249.98
S2	Standard	200	200	150 GB/Partition	512 GB/Partition	36	12	12	\$999.94
S3	Standard	200	200	300 GB/Partition	1 TB/Partition	36	12	12	\$1,999.87
S3HD	High-density	1000	0	300 GB/Partition	1 TB/Partition	36	12	3	\$1,999.87
L1	Storage Optimized	10	10	150 GB/Partition	2 TB/Partition	36	12	12	\$2,856.22
L2	Storage Optimized	10	10	300 GB/Partition	4 TB/Partition	36	12	12	\$5,711.69

Higher storage limits are available for new services in this region at no additional cost.

Azure AI Search: Scale

- Configure the replicas and partitions in scale section.
- Choose 2 replicas for 99.9% availability for read operations, or choose 3 replicas 99.9% availability for read/write operations.
- Choose 1 partitions as default.
- Then, click **Next**.



Microsoft Azure

Search resources, services, and docs (G+)

Home > Microsoft Foundry | AI Search

Create a search service

Basics **Scale** Networking Tags Review + create

SEARCH UNITS ⓘ EST. COST PER MONTH ⓘ

3/₃₆ **\$749.95**

Replicas

Replicas distribute workloads across the service. We guarantee 99.9% availability for read operations with 2 replicas, and for read and write operations with 3 or more replicas. [Learn more about the SLA for Azure AI Search](#)

3

- ✓ 99.9% availability for read operations (choose 2 or more replicas)
- ✓ 99.9% availability for read/write operations (choose 3 or more replicas)

Partitions

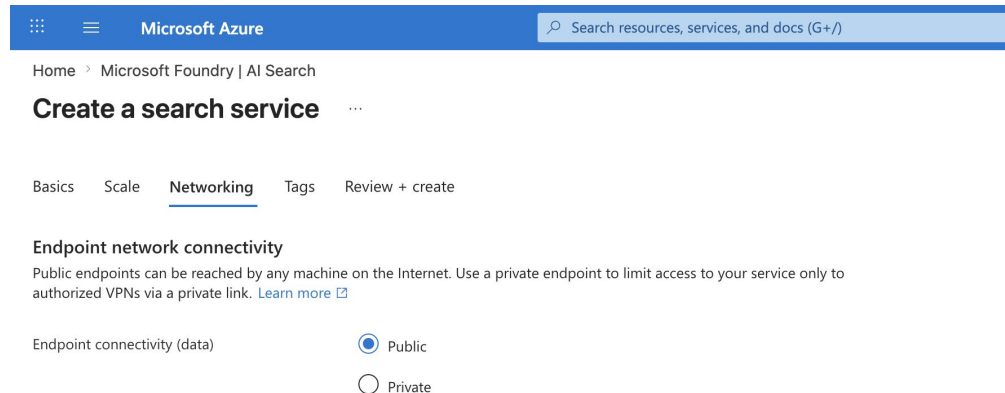
Partitions allow for scaling of document counts as well as faster data ingestion by spanning your index over multiple units.

1

160 GB Storage

Azure AI Search: Network

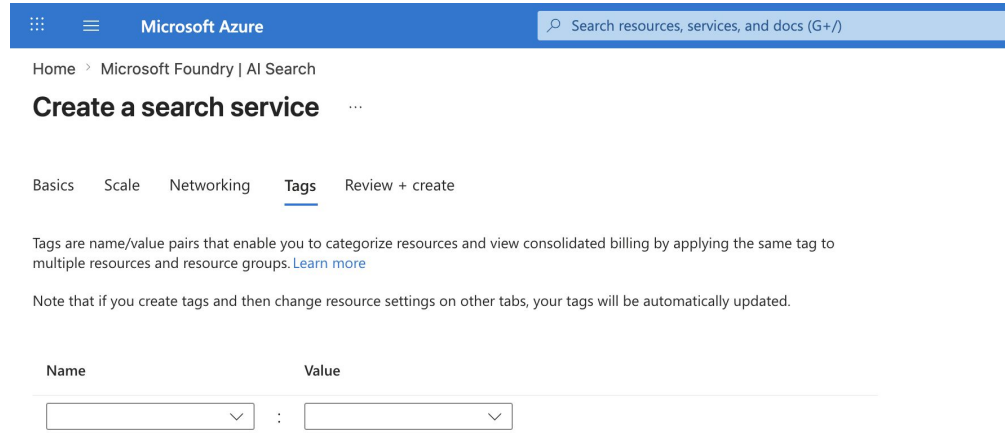
- Configure the network connectivity in networking section.
- Choose public if there are no customize network connectivity for your service.
- Then, click **Next**.



The screenshot shows the Microsoft Azure portal interface. At the top, there's a blue header bar with the Microsoft Azure logo and a search bar. Below the header, the breadcrumb trail reads 'Home > Microsoft Foundry | AI Search'. The main heading is 'Create a search service'. Underneath, there are tabs for 'Basics', 'Scale', 'Networking' (which is selected and underlined), 'Tags', and 'Review + create'. The 'Endpoint network connectivity' section is visible, with a description: 'Public endpoints can be reached by any machine on the Internet. Use a private endpoint to limit access to your service only to authorized VPNs via a private link. [Learn more](#)'. At the bottom, there are two radio button options for 'Endpoint connectivity (data)': 'Public' (which is selected) and 'Private'.

Azure AI Search: Tags

- Configure the name/value in tags section.
- Tags are name/value pairs that enable you to categorize resource.
- Then, click **Next**.



The screenshot shows the 'Create a search service' page in the Microsoft Azure portal. The breadcrumb trail is 'Home > Microsoft Foundry | AI Search'. The page title is 'Create a search service'. Below the title are tabs for 'Basics', 'Scale', 'Networking', 'Tags' (which is selected), and 'Review + create'. A paragraph explains that tags are name/value pairs for categorizing resources and consolidated billing, with a link to 'Learn more'. A note states that tags created on other tabs will be automatically updated. At the bottom, there is a table with two columns: 'Name' and 'Value'. The first row shows two empty dropdown menus separated by a colon, indicating where to enter a tag name and its value.

Microsoft Azure

Search resources, services, and docs (G+)

Home > Microsoft Foundry | AI Search

Create a search service

Basics Scale Networking **Tags** Review + create

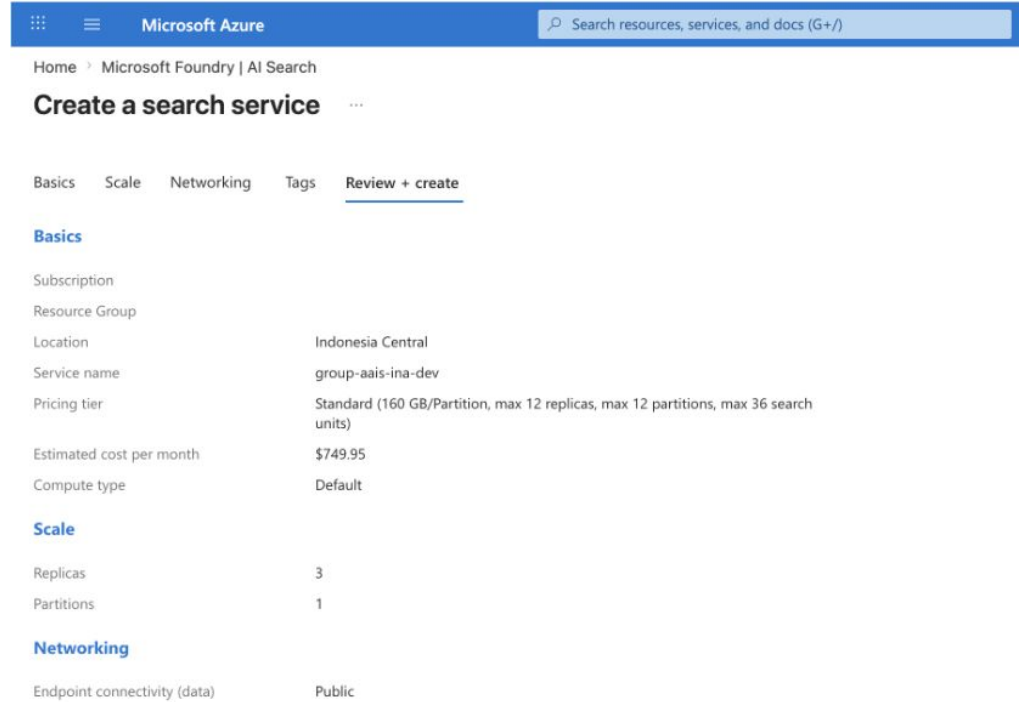
Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name	Value

Azure OpenAI: Review

- Review the basics, scale, and networking settings that have already been configured before submitting.
- If everything is correct, click **Submit**.



The screenshot shows the Microsoft Azure portal interface. At the top, there's a blue header with the Microsoft Azure logo and a search bar. Below the header, the breadcrumb trail reads 'Home > Microsoft Foundry | AI Search'. The main heading is 'Create a search service' with a three-dot menu icon. There are four tabs: 'Basics', 'Scale', 'Networking', and 'Review + create', with the last one being the active tab. Under the 'Basics' tab, there's a list of configuration details:

Subscription	
Resource Group	
Location	Indonesia Central
Service name	group-aais-ina-dev
Pricing tier	Standard (160 GB/Partition, max 12 replicas, max 12 partitions, max 36 search units)
Estimated cost per month	\$749.95
Compute type	Default

Below this, there's a 'Scale' section with two rows:

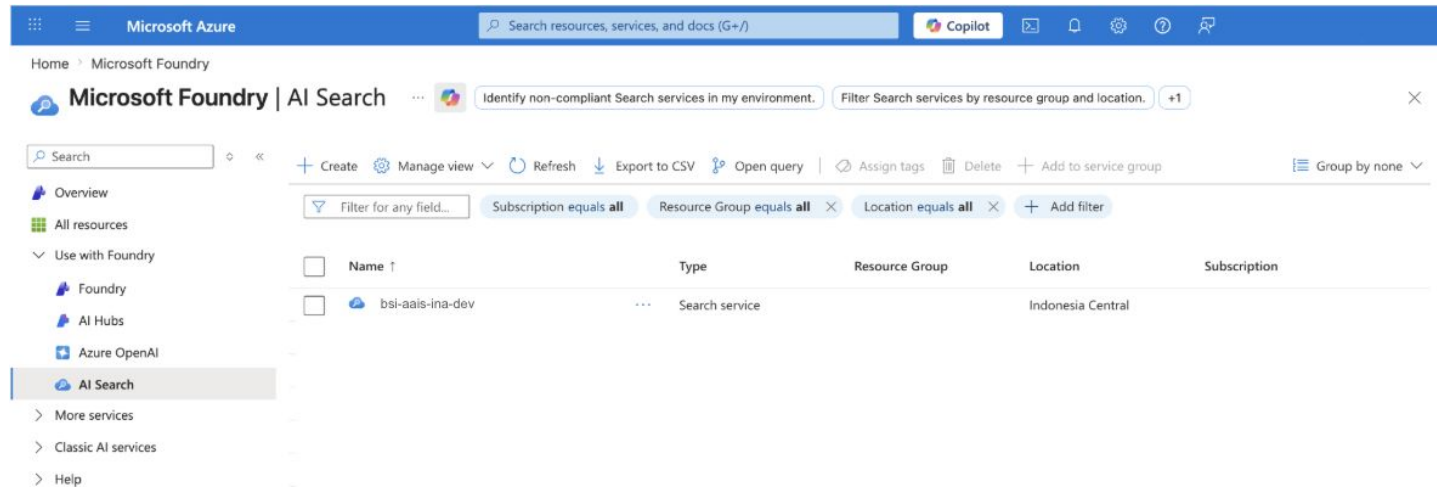
Replicas	3
Partitions	1

Then, there's a 'Networking' section with one row:

Endpoint connectivity (data)	Public
------------------------------	--------

Azure AI Search: Get the API Key

- Open the Microsoft Foundry or Azure AI Search page again.
- Then, click the instance that was created.

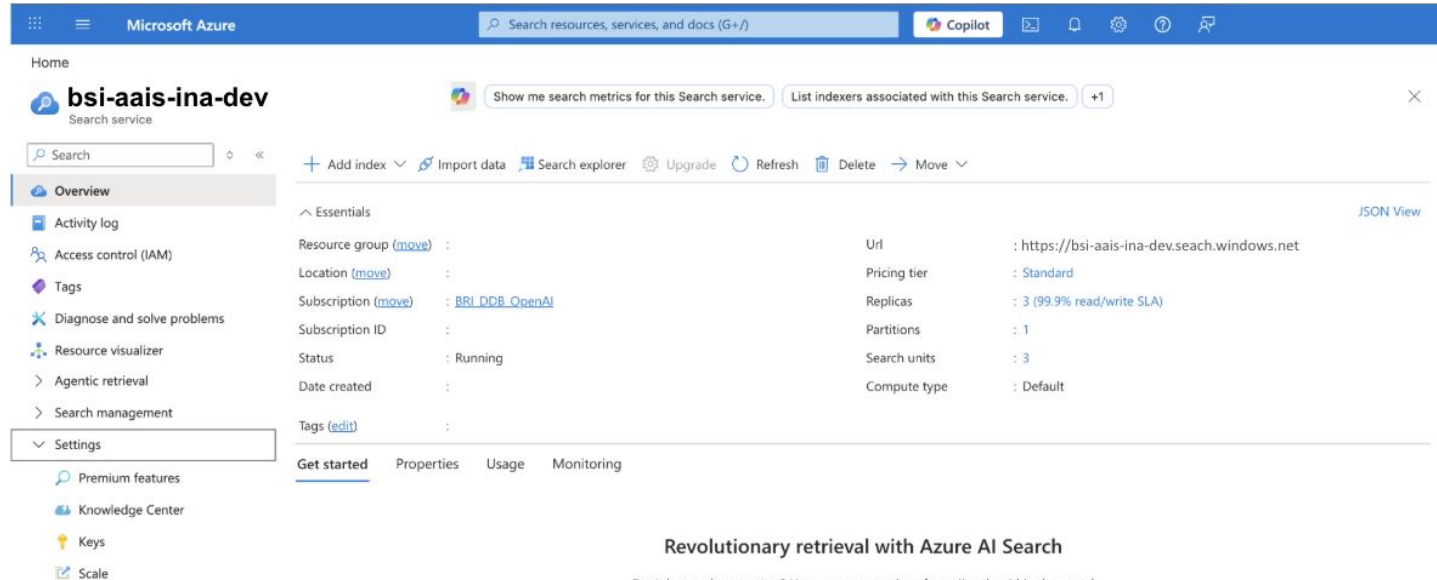


The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and the Copilot icon. The main header indicates the user is in the 'Microsoft Foundry' section, specifically on the 'AI Search' page. A sidebar on the left lists navigation options: Overview, All resources, Use with Foundry (Foundry, AI Hubs, Azure OpenAI, AI Search), More services, Classic AI services, and Help. The AI Search page features a table of search services. The table has columns for Name, Type, Resource Group, Location, and Subscription. One instance is listed: 'bsi-aais-ina-dev' of type 'Search service' in the 'Indonesia Central' location. Above the table, there are filters for Subscription, Resource Group, and Location, all set to 'all'. The table also includes a 'Filter for any field...' dropdown and an 'Add filter' button.

Name ↑	Type	Resource Group	Location	Subscription
bsi-aais-ina-dev	Search service		Indonesia Central	

Azure AI Search: Manage Keys

- It will open the Azure AI Search Instance page.
- Then, click the dropdown **Settings**, then click **Keys**.



The screenshot shows the Azure AI Search portal for the service 'bsi-aais-ina-dev'. The left sidebar contains a navigation menu with the following items: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Agentic retrieval, Search management, and Settings (which is currently selected). Below the Settings item, there are links for Premium features, Knowledge Center, Keys, and Scale. The main content area displays the 'Essentials' section with various properties of the search service, including Resource group, Location, Subscription, Subscription ID, Status, Date created, and Tags. A table of properties is also visible, listing Url, Pricing tier, Replicas, Partitions, Search units, and Compute type. At the bottom of the page, there is a section titled 'Revolutionary retrieval with Azure AI Search' with a link to 'Get started'.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home

bsi-aais-ina-dev
Search service

Show me search metrics for this Search service. List indexers associated with this Search service. +1

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Agentic retrieval

Search management

Settings

Premium features

Knowledge Center

Keys

Scale

Essentials

Resource group (move) : Url : https://bsi-aais-ina-dev.search.windows.net

Location (move) : Pricing tier : Standard

Subscription (move) : BRL DDB OpenAI Replicas : 3 (99.9% read/write SLA)

Subscription ID : Partitions : 1

Status : Running Search units : 3

Date created : Compute type : Default

Tags (edit) :

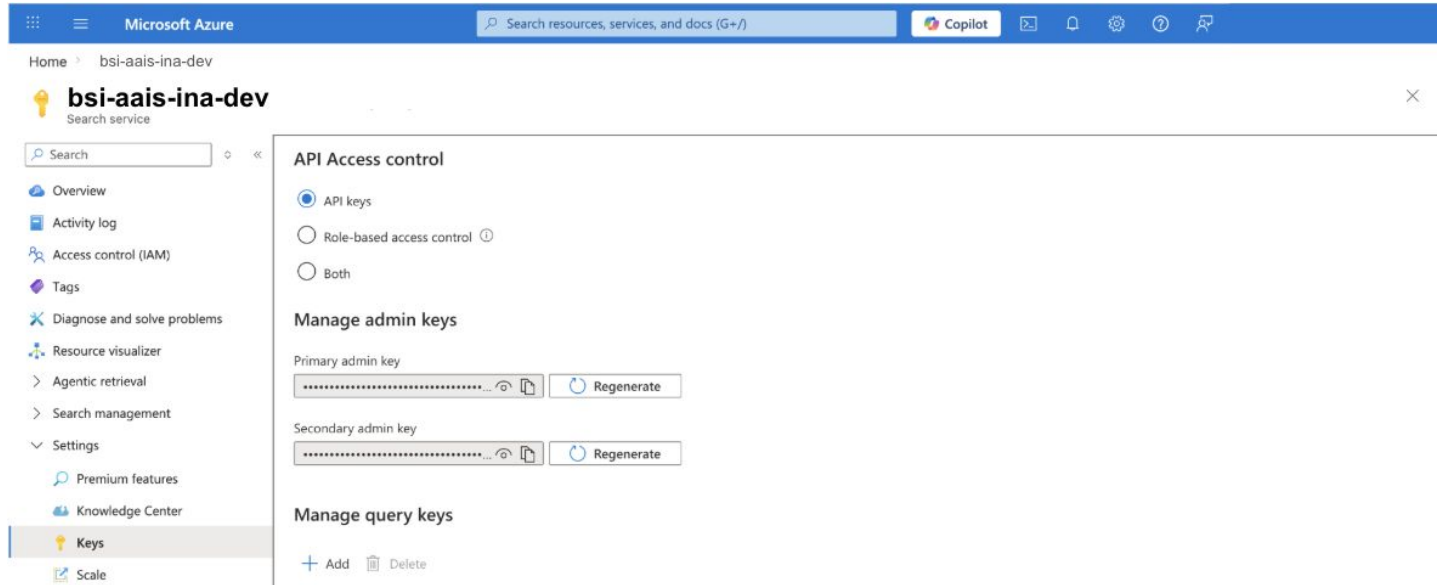
Get started Properties Usage Monitoring

Revolutionary retrieval with Azure AI Search

Don't know where to start? Here are some options from directly within the portal

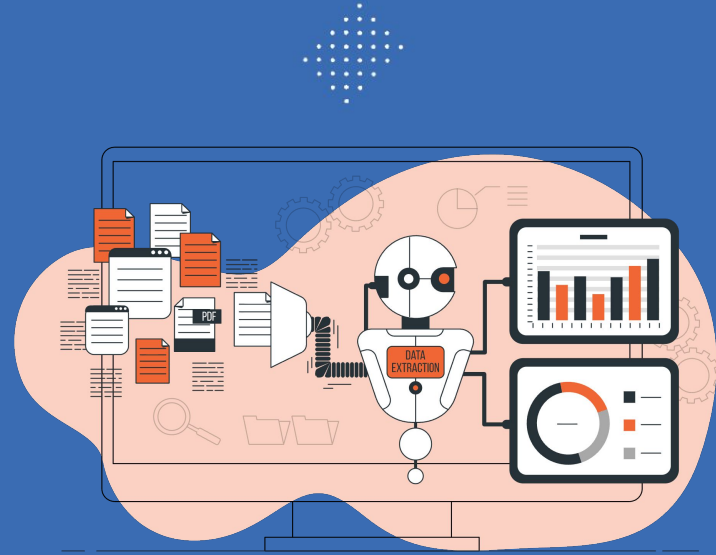
Azure AI Search: Key and Endpoint

- It will open the keys and endpoint section.
- Click the copy icon in primary admin key and secondary admin key for the **API Key**.

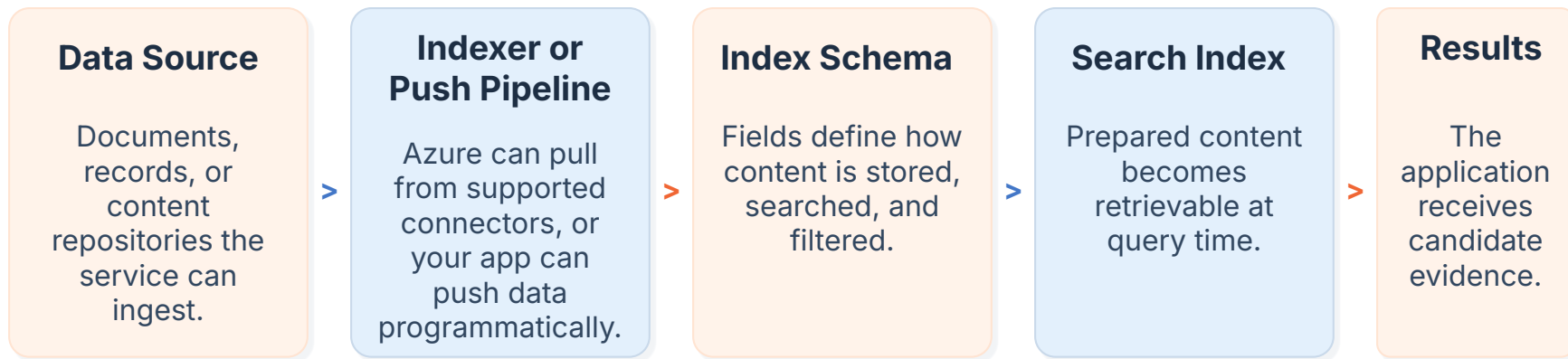


The screenshot displays the Microsoft Azure portal interface. At the top, the navigation bar shows "Microsoft Azure" and a search bar. Below the navigation bar, the breadcrumb trail indicates "Home > bsi-aais-ina-dev". The main content area is titled "bsi-aais-ina-dev" and "Search service". On the left, a sidebar menu lists various options: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Agentic retrieval, Search management, Settings, Premium features, Knowledge Center, Keys (highlighted), and Scale. The main content area is divided into sections: "API Access control" with radio buttons for "API keys" (selected), "Role-based access control", and "Both"; "Manage admin keys" with fields for "Primary admin key" and "Secondary admin key", each with a "Regenerate" button; and "Manage query keys" with "Add" and "Delete" buttons.

Azure AI Search Data Sources & Indexing



Azure AI Search Indexing Flow



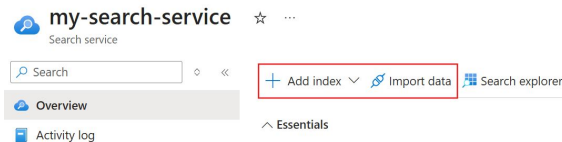
Indexing is not just setup work. Field design and ingestion choices directly shape retrieval quality later, which means they also shape the quality of grounded model answers.

Index Schema and Field Design

Why schema design matters

A knowledge base is only as searchable as the fields you design into the index. Titles, body text, category, document source, and timestamps are not busywork. They determine what the service can retrieve, filter, or rank later.

Index Creation Screen



Use the field definitions screen to connect abstract schema ideas to concrete search design.

1

Title and Summary

Create a human-readable entry point for retrieval results.

2

Body Text

Carry the main searchable evidence that later supports grounding.

3

Category and Filters

Support grouping, filtering, and downstream query control.

4

Source and Timestamp

Help explain provenance, freshness, and document context.

Indexer Workflow

Import Data / Indexer Flow

- > Security
- > Development
- > Monitoring
- > Samples
- > Responsible AI
- > References
- > Resources

- **Raw content indexing (built-in or custom):** Extracts raw vector content for full-text search scenarios. It also extracts raw vector content for vector search (for example, vectors in an Azure SQL database or Azure Cosmos DB collection). In this workflow, indexing occurs only over existing content that you provide.
- **Skills-based indexing:** Extends indexing through built-in or custom skills that generate new searchable content. For example, you can add integrated machine learning for analysis over images and unstructured text, extracting or inferring text and structure. You can also use skills to chunk and vectorize content from text and images. Skills-based indexing generates new content that doesn't exist in your external data source. New content becomes part of your index when you add fields to the index schema that accept incoming data. To learn more, see [AI enrichment in Azure AI Search](#).

Prerequisites

- A [supported data source](#) that contains the content you want to ingest.
- A [data source](#) definition that sets up a connection to external data.
- A [search index](#) that can accept incoming data.
- Be under the [maximum limits](#) for your service tier. The Free tier allows three objects of each type and 1-3 minutes of indexer processing, or 3-10 minutes if there's a skillset.

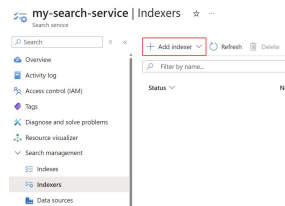
Indexer patterns

When you create an indexer, the definition follows one of two patterns: *content-based indexing* or *skills-based indexing*. The patterns are similar, but skills-based indexing includes extra properties.

Indexer example for content-based indexing

This workflow groups index, indexer, and ingestion setup into one retrieval-preparation experience.

Create Indexer



Indexers automate the pull process from supported data sources.

1

Open the search service workflow

Index creation and ingestion setup are connected in one workflow.

2

Automation boundary

Indexers are convenient, but retrieval quality still depends on good source data and schema design.

Data Sources and Connector Strategy

Core design point

Azure AI Search does not ingest content from anywhere without a plan. Connector choice is part of architecture: teams must know where content lives, whether a supported connector exists, and whether a pull-based workflow fits the use case.

1 Where does the content live?

Connector support starts with the actual content location.

2 Is pull-based ingestion enough?

If yes, indexers can be a clean fit. If not, push pipelines may be better.

Data Sources Gallery

- What is an indexer?
- > Create and manage
- Index from data sources
 - Supported data sources
 - > Azure Storage
 - > Azure Cosmos DB
 - Azure DB for MySQL
 - > Azure SQL
 - OneLake
 - SharePoint in Microsoft 365
 - > Index whole files
 - > Troubleshooting
 - > Tutorials
 - > Logic Apps workflows
 - > Skillssets (AI enrichment)
 - > Retrieval
 - > Relevance and ranking
 - > Agentic retrieval (RAG)
 - > Security
 - > Development
 - > Monitoring
 - > Examples

Pull in content from other Azure services using indexers and the following data source connectors.

Azure Blob Storage

By [Azure AI Search](#)

Extract blob metadata and content, serialized into JSON documents, and imported into a search index as search documents. Set properties in both data source and indexer definitions to optimize for various blob content types. Change detection is supported automatically.

[More details](#)



Azure Table Storage

By [Azure AI Search](#)

Extract rows from an Azure Table, serialized into JSON documents, and imported into a search index as search documents.

[More details](#)



Azure Data Lake Storage Gen2

By [Azure AI Search](#)

Connect to Azure Storage through Azure Data Lake Storage Gen2 to extract content from a hierarchy of directories and nested subdirectories.

[More details](#)



Connector support is part of solution design because it determines how content reaches the index.

Pull vs Push Ingestion

Pull with indexers

Best when the source type is supported and the schedule is predictable.

Advantage: less custom ingestion plumbing.

Constraint: you still need good source data, a fitting connector, and a schema that supports retrieval quality.

Push programmatically

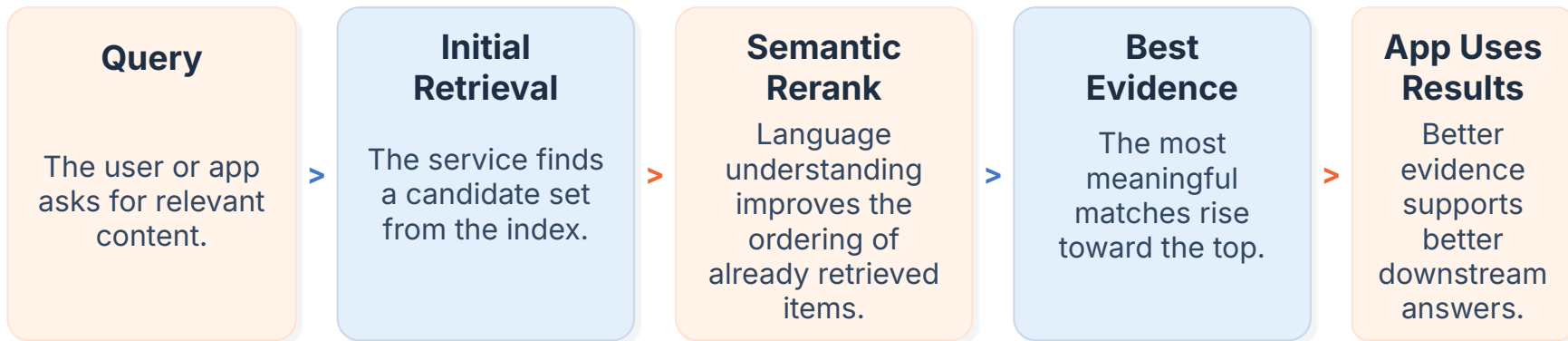
Best when the application needs more control over ingestion timing or transformation.

Advantage: more flexible pipeline control.

Constraint: your team owns more of the ingestion logic and quality discipline.

Key takeaway: Azure AI Search supports both preparation styles. The right choice depends on the source system and the operational control the team needs.

Semantic Ranking



Semantic Guidance

Semantic ranking sits on top of the retrieval pipeline. It is additive, not a substitute for indexing discipline.

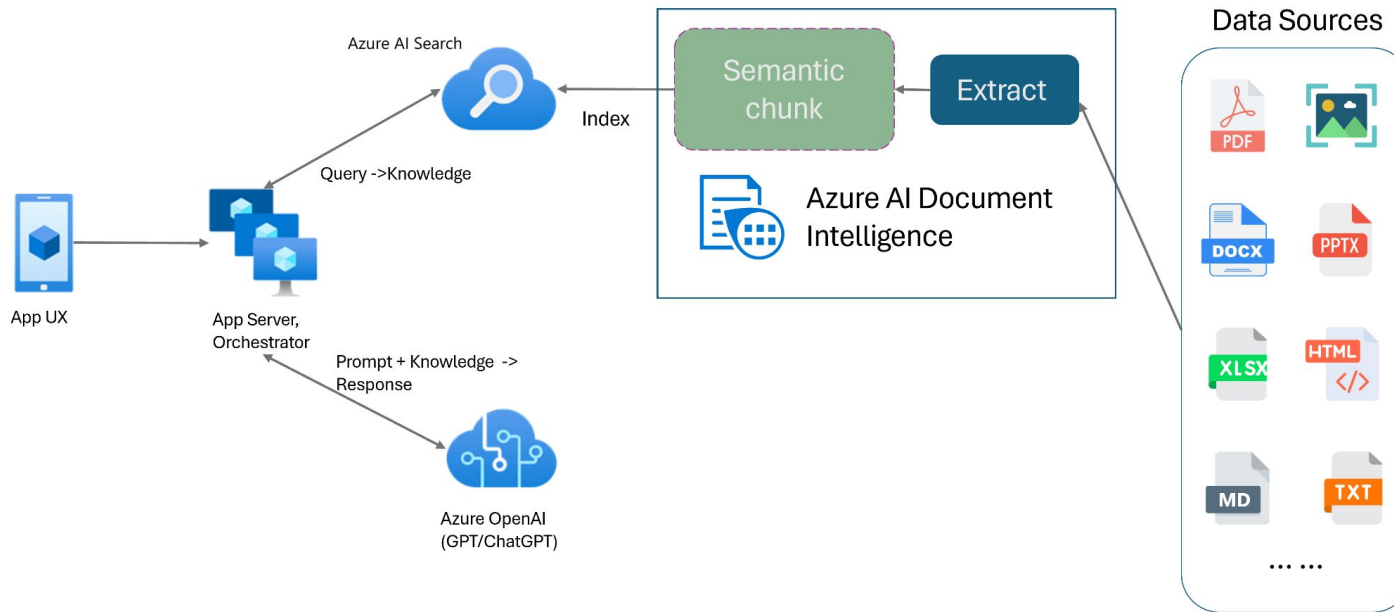
What semantic ranking does not do

Semantic ranking does not replace indexing. It does not rescue weak source data or a weak schema. It improves ordering after initial retrieval, which matters because retrieved evidence strongly shapes answer quality in RAG systems.

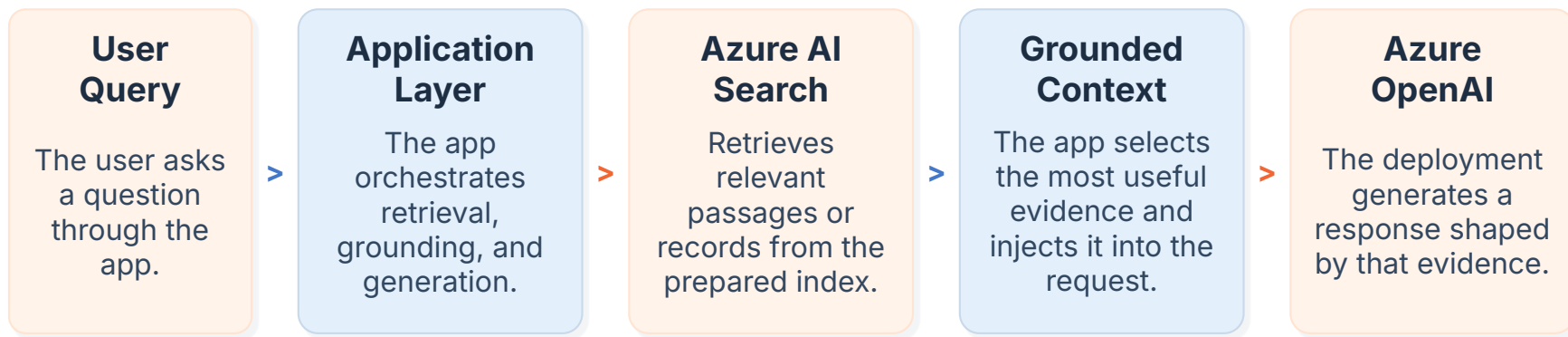
Building Retrieval Augmented Generation



AI Search RAG Architecture



Integrate RAG with AI Search



Sequence matters: retrieval happens before generation. Azure AI Search is the retrieval layer, Azure OpenAI is the generation layer, and the application stitches them together into a real solution.

Guided Hands-on



IT Helpdesk Knowledge Base

Learning Goals

Participants will learn to securely connect to Azure OpenAI and Azure AI Search, design a simple index schema, ingest documents using the push API, and build a complete end-to-end RAG pipeline that retrieves relevant content, grounds the LLM, and produces cited answers, including handling out-of-scope questions gracefully.

Setup

Participants can use:

- Google Colaboratory:
[Session 35 Hands-on_Azure OpenAI and AI Search.ipynb](#)
- Azure OpenAI Keys and Endpoint
- Azure AI Search Keys and Endpoint

IT Helpdesk Knowledge Base



Use Case

IT Helpdesk Knowledge Base Assistant. A small RAG assistant that answers internal IT questions for BSI employees (VPN, password reset, MFA, approved software, etc.) using ~10 curated knowledge-base articles.



Tasks

- Load Azure credentials from Colab Secrets
- Connect to Azure OpenAI and run a smoke test
- Design and create a simple AI Search index
- Push 10 helpdesk articles into the index
- Run a keyword search and inspect scores
- Build the RAG pipeline (retrieve → ground → generate)
- Test with in-scope and out-of-scope questions

Assignment



Vehicle Service Manual Assistant

Learning Goals

Participants will architect a production-style index with semantic ranking, implement and compare three retrieval strategies (keyword, semantic, and filtered), and build a multi-turn RAG chatbot with conversation history. The assignment ends with a side-by-side evaluation that pushes participants to reason critically about when semantic ranking helps and when schema design matters more.

Setup

Participants can use:

- Google Colaboratory:
[Session 35 Assignment_Azure OpenAI and AI Search.ipynb](#)
- Azure OpenAI Keys and Endpoint
- Azure AI Search Keys and Endpoint

Vehicle Service Manual Assistant



Use Case

Vehicle Service Manual Assistant. Internal AI assistant for Mitsubishi Indonesia After-Sales service advisors. Service advisors ask natural-language questions about common vehicle issues; your RAG system retrieves from a richer service manual knowledge base (engine, battery, transmission, brakes, HVAC, electrical, fuel, suspension, cooling, general) and produces grounded, cited answers with category filtering and multi-turn conversation memory.



Tasks

- Load credentials and connect to Azure OpenAI
- Design an 8-field index with semantic configuration
- Upload 15 service manual articles
- Implement keyword, semantic, and filtered search
- Build a multi-turn RAG chatbot with citations
- Compare keyword vs semantic on 5 test questions
- Write a short analysis of the results



Thank You